Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Software engineer with 6+ years of professional experience building production AI systems, LLM applications, and distributed backend infrastructure. At Intuit, designing and deploying agentic AI workflows using the Anthropic Messages API and Agent SDK -- RAG architectures, prompt engineering, agentic orchestration, and context-augmented generation in production financial data services. Built LeaseI, an 8-agent LangGraph system (93.6% precision, 87.9% F1 across 1,240 test cases) demonstrating end-to-end RAG pipeline design, vector retrieval, and human-in-the-loop escalation. At Entrata (2018--2024), led architecture and design of eligibility decisioning systems, ETL pipelines processing 1.6M+ records, and distributed async infrastructure at scale -- mentoring junior engineers and driving technical direction across teams. Strong Python, SQL, AWS, Docker, Kubernetes, Terraform. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / FastAPI / SQL | LLM Application Development | RAG Architecture / Vector Databases | Prompt & Context Engineering | Agentic Orchestration (LangGraph) | Messages API / Agent SDK | Fine-tuning / Few-shot Learning | ML Pipelines (scikit-learn, XGBoost) | Distributed Systems / Microservices | AWS (Lambda, S3, RDS, SQS) | Kubernetes / Terraform (IaC) | Async Processing (RabbitMQ) | CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry)

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- LLM Applications & AI Infrastructure -- Virginia, USA

- Design and build **LLM application infrastructure** using the Anthropic Messages API and Agent SDK: RAG pipelines, embedding-based retrieval, prompt engineering, tool-use patterns, and context-augmented generation for production financial data services.
- Implement **agentic AI workflows** with multi-step orchestration, chain-of-thought reasoning, and guardrails; build golden-dataset eval harnesses to validate AI output quality before every release.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through prediction APIs; deploy on AWS via Docker and GitHub Actions CI/CD; instrument with Prometheus and OpenTelemetry.
- Prototype new AI capabilities (few-shot learning, fine-tuning approaches), evaluate build vs. buy tradeoffs, and drive responsible AI practices including bias review and accuracy validation.

Entrata -- Residential & Affordable Housing SaaS

Sep 2018 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- **Led architecture and design** of the eligibility decisioning engine -- AMI threshold calculations, income verification rules, HUD/LIHTC compliance routing across 100K+ units. Owned system design from requirements through production, zero compliance violations over 6 years.
- Designed **distributed async infrastructure** (RabbitMQ) for real-time eligibility and rent updates across thousands of tenant records -- idempotency guarantees, retry logic, dead-letter handling, and scalability patterns for a high-throughput multi-tenant system.
- Owned **ETL pipelines processing 1.6M+ records**: federal HUD API ingestion, transformation into ~180,000 domain records, validation, reconciliation, and rollback at each release cycle.
- Optimized 200+ SQL queries via execution plan analysis (30% performance gain across 1M+ records); designed multi-schema PostgreSQL/Oracle data models for long-term scalability.
- Led AWS S3 migration with PHI encryption (Libsodium); designed Docker microservices architecture (50% better scalability); deployed via Terraform IaC.
- **Mentored junior engineers** on distributed systems, data pipeline design, and engineering quality; reviewed 300+ PRs; drove technical standards across the team.

PROJECTS

Leasel -- RAG + Agentic AI System

Python · LangGraph · RAG · Vector Search · FastAPI · Evals

8-agent LangGraph orchestration system: RAG retrieval pipeline with embedding-based vector search, context-augmented generation, confidence scoring, and human escalation for low-confidence outputs. Chain-of-thought reasoning at each agent step. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- LLM Inference & Evaluation Infrastructure

Python · FastAPI · Provider Routing · Cost Management · CI/CD

15-endpoint FastAPI service routing across five LLM/ML inference backends with cost-aware model selection. Scoring engine evaluates output quality on five metrics; automated CI regression against golden datasets. Addresses AI infrastructure problems: observability, cost management, and flexible model adoption at scale.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical NLP Application

Python · Flask · NLP · REST APIs · RBAC · Docker

RESTful API translating clinical natural language queries into FHIR data calls using LLM-based NLP. Fine-grained access control, containerized deployment, and automated testing -- regulated-domain LLM application pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017